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An examination into Star Ratings based on Official Yelp Dataset

By Soraya Shirini, Olive Edeko, Gnanesh Reddy, Ariane Damian and Serena Emedeke

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Student Name	**Student ID**	**Signature**	**Date**
Soraya Shirini	A00090611	Soraya Shirini	02/12/2025
Olive Edeko	A21381813	Olive Edeko	02/12/2025
Gnanesh Reddy	A00091407	R.Gnanesh	02/12/2025
Ariane Damian	A00094109	A. Damian	02/12/2025
Serena Emedeke	A0092238	S. Emedeke	02/12/2025

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1. INTRODUCTION

Reputation is a direct reflection of a business' performance. It serves as an indicator of their image, reliability, and quality, all of which combine to influence consumer behaviour (Porter, 1974). With the proliferation of technology, online review websites have become an essential source for determining a business' reputation (Wu et al., 2015). Yelp is one of the most established consumer websites for both publishing and searching information about the quality of a business' products/services (Fogel & Zachariah, 2017; Salehi-Esfahani & Kang, 2019). Yelp also provides multiple services to local businesses to boost visibility, leading to an increase in consumer interest and a direct growth in revenue. One of key methods of achieving this is through 'Star Rating'. Evidence has demonstrated that even an increase by one star rating results in a substantial increase in revenue of up to 9% (Luca, 2016).

In this report, we endeavour to use an official dataset provided by Yelp in order to propose a new business model using 'Star Rating' as the target variable. One of Yelp's biggest advantages is their vast coverage across multiple industries; as a company built for advertising, it is highly multifaceted in that it covers not just basic marketing but also e-commerce and data. This proposed business model leverages this versatility in order to offer a niche service in analytics through a 'Star Rating Prediction'. All businesses, including those in hospitality, healthcare, finance and more can subsequently avail of this service in order to improve their own services.

A challenge based learning (CBL) approach is applied to connect technical work to real business problems. The platform must recommend reliable business, but many have few or inconsistent reviews. Predicting star ratings helps Yelp estimate business quality, improve ranking and recommendation decisions,and identify businesses to highlight in "Best of Yelp" lists. Within this CBL framework, the elements of the challenge are summarized in *Table 1.1* below.

Big question:	How can a platform like Yelp estimate quality in a reliable way
The challenge:	Predict the star rating of a business using only business level information such as attributes, category, location, review count.
Operational decision:	Yelp often highlights or promotes certain businesses in search results and "Best of Yelp " lists. Should this business be shown or recommended more to users?
Key Performance Indicator (KPI):	The main KPI in this project is the predicted star rating (a continuous value from 1 to 5). A higher predicted rating suggests a higher-quality business.
Stakeholders:	1.) Business owners, who want more visibility and 2.) Yelp users, who rely on accurate reccomendations

Table 1.1 Summary of challenges/characteristics identified through CBL

Exploratory Data Analysis (EDA) is conducted prior to the application of statistical methods. Linear regression, random forest regression and gradient boosting regression were utilised to assess the relationships between key attributes and star rating. Unfortunately, several limitations exist within the dataset. Recommendations are made based on primary findings and further inferences based on available literature.

2. METHODOLOGY/MODELLING

2.1 Data preparation

The raw dataset consists of approximately 150,000 rows and 14 columns. Missing and null values were removed. Columns *'hours'*, *'address'* and *'postal_code'* were excluded as they lack relevance in measuring the target variable. Category and Attributes were split into primary categories and primary attributes based on the frequency of each component; 12 key categories and 33 key attributes were identified. Post data preparation, the final dataset comprises 15,0243 rows and 56 columns. Data specifications are outlined in *Appendices: Table 2.1*

2.2 Modelling techniques

Three modelling techniques were used to predict 'Star Rating'; Linear regression, Random Forest regression and Gradient boosting regression. Linear regression was first applied to achieve a general understanding of variables that may influence 'Star Rating'. It serves as a foundation for further statistical modelling and is proven to be highly effective when implemented early in the analysis process (James et al., 2023).

However it introduces a set of disadvantages i.e. it requires assumptions that are not always applicable to large, complex datasets. Particularly, it assumes a relationship of linearity between its variables (Ogunleye, 2022). Predictive analysis often involves non-linear relationships. In such cases, Random Forest Regressors should be utilised as it makes no assumptions regarding its data's variables (*What Are the Advantages and Disadvantages of Random Forest?*). Additionally, it is less sensitive to outliers, missing data and other features that generate noise, eliminating the need to handle/transform the data before modelling.

Gradient Boosting Regressor was implemented as the third, advanced modelling technique to maximise the accuracy of the star rating prediction. It performs exceptionally well in situations where high accuracy is prioritised (Kalusivalingam et al., 2022). This is achieved through sequential and iterative correction of errors in previous models, until a stopping criteria is met, reducing systematic errors as well as fine-tuning predictive accuracy (Muneer, 2024). Combined with linear regression and random forest regression, CBL is applied to identify the most robust and precise method of measuring 'Star Rating' in order to support the new proposed business model.

3. ANALYSIS/DISCUSSION

3.1 EDA Visualisation

In order to better understand the dataset, EDA was conducted post data processing. EDA is an especially crucial step prior to modelling; it allows one to familiarise themselves with the dataset, and identify patterns and hidden relationships between variables (Cox, 2017). Subsequently, it aids in selecting appropriate statistical models for the purpose of answering the business question/problem. Below are a set of visualisations executed in Python, with each code outlined in the provided Jupyter notebook.

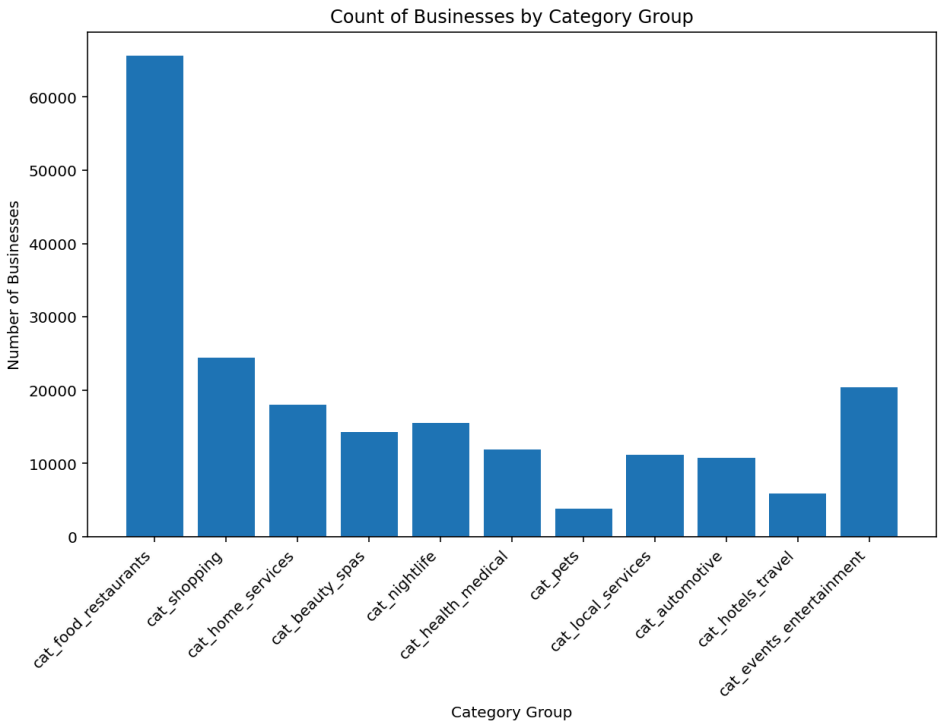

Fig 3.11 Number of business by category group

Fig 3.11 exhibits the number of businesses belonging to each of the main category groups. The largest category identified is *Food & Restaurants*, with approximately 65,000 businesses. A significant gap is observed between this category and the next most prevalent sectors e.g, *Shopping* and *Home Services*.

The market appears to be heavily saturated with food businesses, demonstrating a highly competitive industry. Gaining visibility in such conditions is difficult especially for smaller businesses, whose capacity for standing out is not equatable to industry giants with access to

more revenue, connections and spread. In contrast, categories such as *Pets* or *Hotels/Travels* have less competition, which offers more opportunities to build customer relationships.

In both scenarios, particularly in the first one, Yelp's 'Star Rating Prediction' can be leveraged to aid smaller and more niche businesses to identify characteristics that drive strong, positive performances in successful businesses. The 'Star Rating Prediction' can allow these unknown businesses to first, forecast their own performance and second, to compare against established examples and help shift focus towards adopting recognised strategies. More importantly, it can examine areas less utilised by other businesses and confirm if there is potential to develop that area in order to stand out from the competition.

In terms of using this feature to model 'Star Rating Prediction', there are some notable caveats. As the overall distribution of business is uneven i.e. some business types are much more common in the dataset compared to others, modelling may not be entirely accurate for categories with considerably less data compared to those that dominate the market.

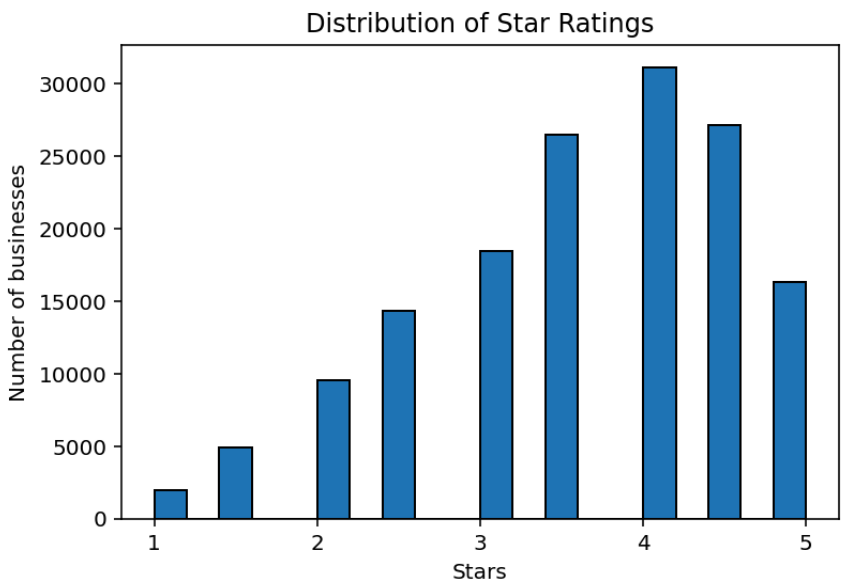

Fig 3.12 Distribution of star ratings

Most businesses are rated between 3 and 4.5 stars with 4 stars being the most common according to ***Fig 3.12***. Similar to the distribution of categories across the data set, some star groups contain a disproportionate number of data compared to other Stars. This can result in

difficulties with modelling for the same reasons - as the model has fewer examples of poorly rated business to learn from, prediction can be less accurate.

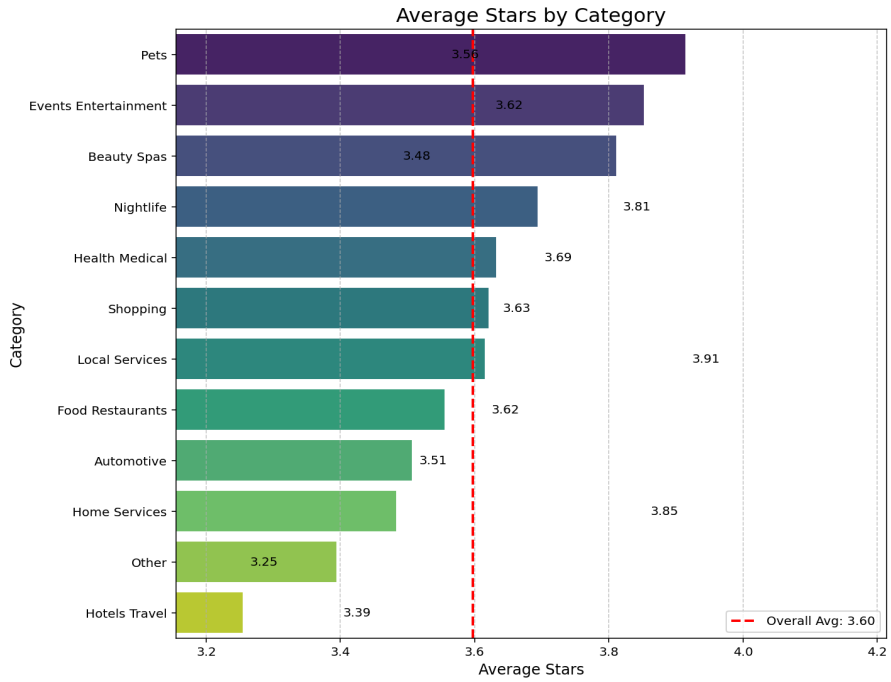

Fig 3.13 Average star ratings across/within categories

Fig 3.13 determines if specific business categories obtain higher star ratings comparative to others. The horizontal dash represents the overall average star rating of 3.6 stars across all business categories. While there are some categories above the average star rating (e.g cat_pets), most categories cluster in and around the average, implying that the individual business is the biggest determining factor of star rating and not the business category it operates in.

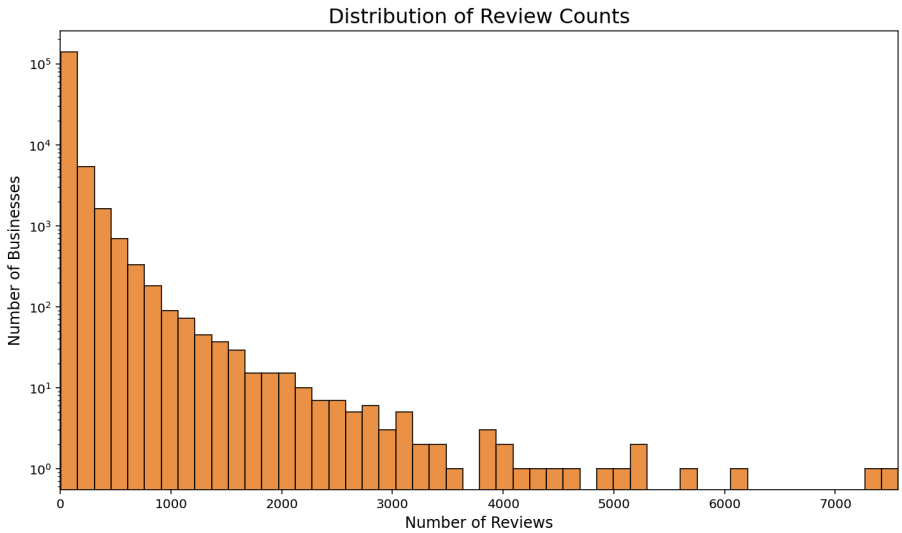

Fig 3.14: Distribution of reviews across a number of businesses

Another notable feature that can affect how Yelp's model predicts star ratings is the distribution of reviews across the vast number of businesses. ***Fig 3.14 above*** depicts how most businesses receive very few reviews, while a small number of businesses have an extremely high review count. The plot uses a log scale because review numbers vary widely.

Additionally, when attempting to correlate the number of reviews against star rating, it was found that the former has no massive impact on the star rating given. However, star rating remains strongly affected by the type and characteristics of a business. This is visualized as per ***Fig 3.15 below*** which portrays the wide variety of star ratings in spite of the number of reviews of the business.

Both ***Fig 3.14*** and ***Fig 3.15*** highlights an imbalance that is crucial for modelling, demonstrating how review_count becomes a strong but uneven predictor of star ratings.

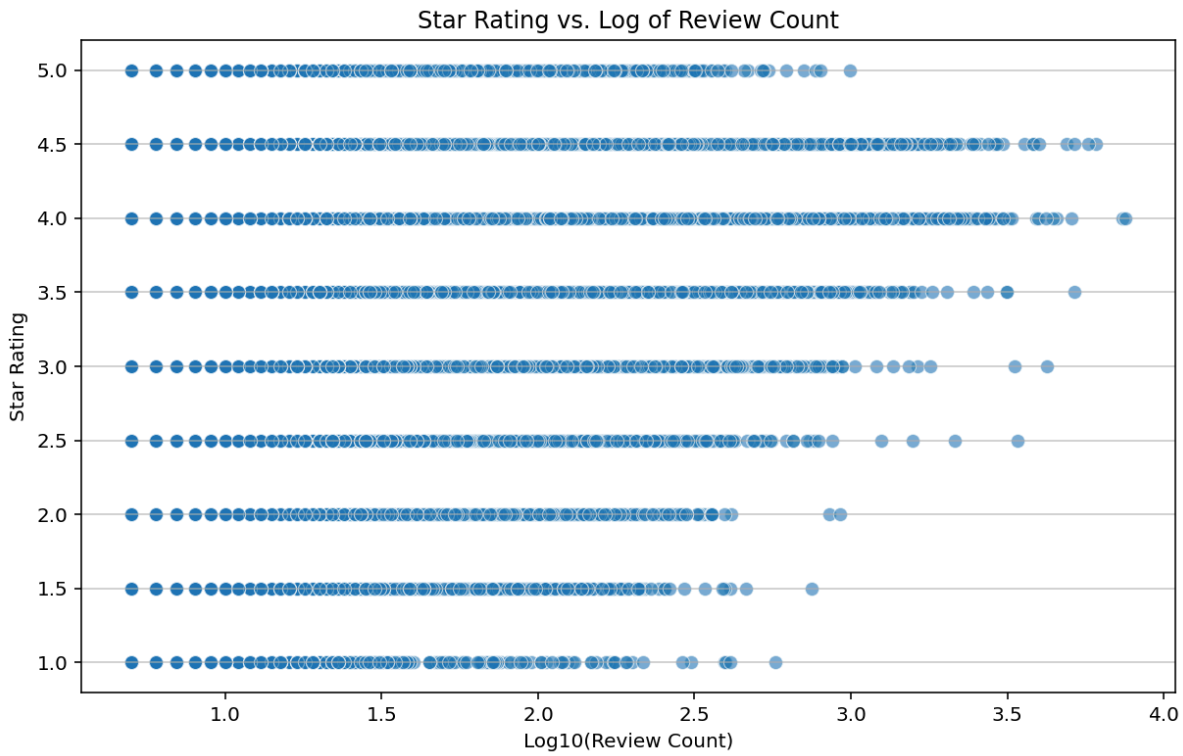

Fig 3.15: Distribution of star ratings against review counts

While the number of reviews may not be directly correlated, one aspect that Yelp can look into are attributes or characteristics identified within those reviews. Particularly, they can measure the impact of business attributes on star rating.

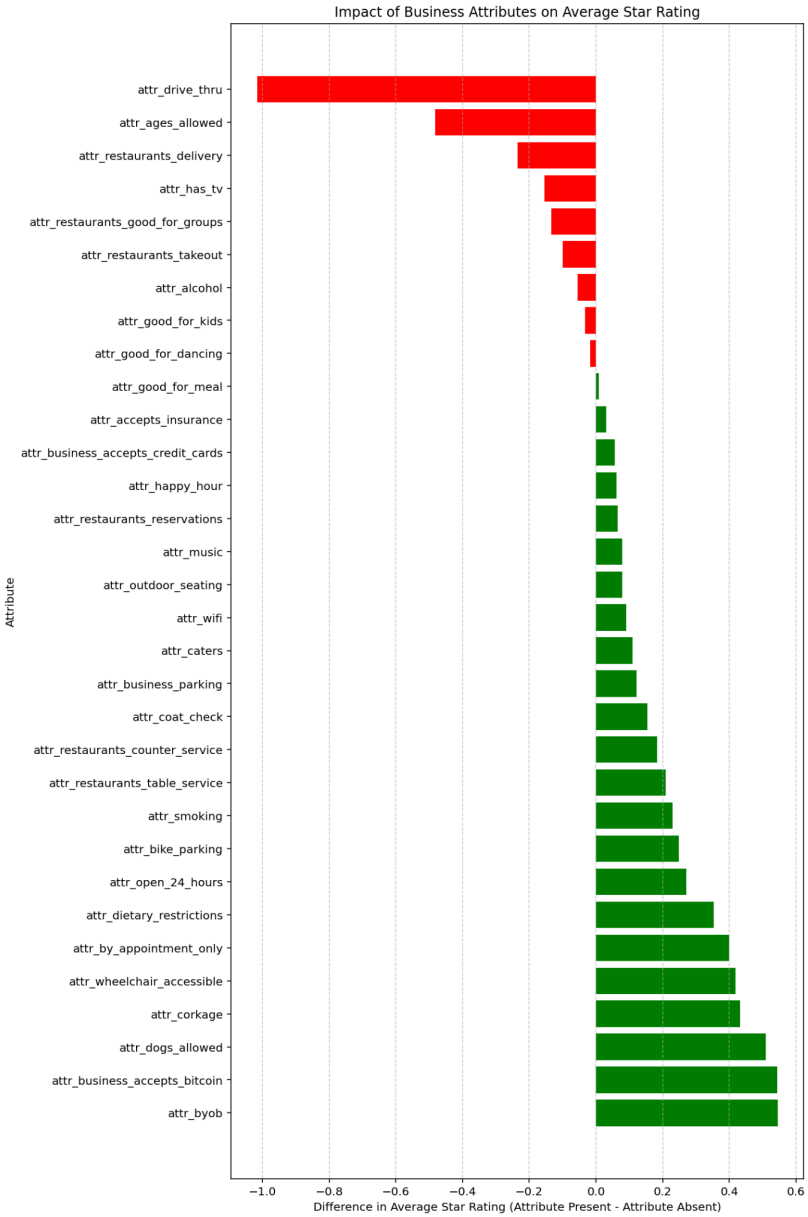

Fig 3.16: Impact of Business attributes on star rating

Fig 3.16 shows the difference in average star rating if an attribute is present i.e. does the business accept bitcoin or not, is there bike parking available, etc. Green denotes attributes that are associated with positive star rating while red indicates the opposite. ***Fig 3.15*** demonstrates that the presence of most attributes are encouraged, however whether or not this can predict the star rating of a business is a question for the machine model to uncover.

For Yelp, ***Fig 3.14*** which shows the lack of reviews, and ***Fig 3.16*** which attempts to examine characteristics associated with a positive star rating presents two opportunities; first, a lack of

reviews can indicate low visibility or perhaps low/neutral customer satisfaction. If the issue was the former, Yelp can utilise this data to provide advertising/marketing services to businesses facing this problem. If the issue was low/neutral customer satisfaction, Yelp can provide evidence as to why struggling businesses are being outperformed.

Secondly, if they were to present these avenues for growth to local business and were successful in increasing customer engagement, the sample size for which Yelp develops their 'Star Rating Prediction' model becomes greater, resulting in a more precise and accurate model.

3.2 Regression Analysis

Regression analysis was conducted on two datasets to assess whether a business's star rating can be predicted using categories or attributes, alongside location and review activity. The aim was to simplify the models and evaluate the strength of these relationships. R^2 values and error metrics, the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), are used to assess the model, and are summarized in ***Table 2.2*** below

	Modelling	**R squared**	**MAE**	**RMSE**
Categories	Linear regression (LR)	0.045	0.77	0.95
	Random Forest Regressor (RFR)	0.093	0.74	0.93
	Gradient Boosting Regressor (GBR)	0.086	0.74	0.87
Attributes	Linear regression (LR)	0.13	0.73	0.91
	Random Forest Regressor (RFR)	0.19	0.69	0.88
	Gradient Boosting Regressor (GBR)	0.17	0.71	0.84

Table 3.2 Summary of key metrics for assessing model performance in both categories and attributes

LR attempts to explore possible linear relationships from categories and attributes, alongside location and review activity; it performed the worst for both, with the lowest R^2, and the

highest MAE, and RMSE values. R^2 explains the variance (ideally high) while MAE and RMSE demonstrate the model's accuracy and deviation from actual values (ideally low). Attributes is slightly better at predicting rating compared to categories, however overall, the model failed to capture complex patterns, highlighting that predicted ratings deviate from actual ratings by nearly 1 star on average.

RFR performs slightly better, however it remains a weak predictive model overall. Both error metrics in categories and attributes deviate by almost a full star on average, reflecting significant prediction errors.

GBR is the most advanced tree-based model, and shows slight differences between categories and attributes. In categories, it performs worse in variance compared to RFR but edges out slightly in error metrics. In attributes, GBR slightly outperforms RFR only in RMSE. Considering all these factors, RFR appears to be the most robust model, although the differences are marginal. No model excels in predictive accuracy. All three models are severely limited by the available, business level only data, and richer features such as review text and price information would likely improve predictions.

4. CONCLUSION

Reputation is the backbone of a businesses' success, and visibility is paramount in order to increase and maintain reputation. Yelp provides a platform for local businesses to connect with consumers through advertising/reviews, however there's a potential market for analytics. Star Prediction can be offered to businesses who want to improve their reputation and perception. Yelp should develop this as a niche, premium service that business can avail of; it should provide a forecast of a business' performance according to customer satisfaction, reasons for apparent disrepute and recommendations for improvement. Additionally, this data should be utilised internally by Yelp to improve operational decisions. Yelp often highlights or promotes certain businesses in search results and "Best of Yelp " lists (*Yelp Business Model - How Does Yelp Make Money?*, 2022) . By predicting the star rating of certain businesses, Yelp can avoid promoting those with ill reputation, cementing their reliability and reputation as a consumer-interactive platform.

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APPENDICES

Column	Non-Null	Count	Dtype
business_id	150243	non-null	object
name	150243	non-null	object
city	150243	non-null	object
state	150243	non-null	object
latitude	150243	non-null	float64
longitude	150243	non-null	float64
stars	150243	non-null	float64
review_count	150243	non-null	int64
is_open	150243	non-null	int64
categories	150243	non-null	object
cat_food_restaurants	150243	non-null	int64
cat_shopping	150243	non-null	int64
cat_home_services	150243	non-null	int64
cat_beauty_spas	150243	non-null	int64
cat_nightlife	150243	non-null	int64
cat_health_medical	150243	non-null	int64
cat_pets	150243	non-null	int64
cat_local_services	150243	non-null	int64
cat_automotive	150243	non-null	int64
cat_events_entertainment	150243	non-null	int64
cat_hotels_travel	150243	non-null	int64
cat_other	150243	non-null	int64
attributes	136601	non-null	object
attr_business_accepts_credit_cards	150243	non-null	int64
attr_business_parking	150243	non-null	int64

attr_bike_parking	150243	non-null	int64
attr_restaurants_takeout	150243	non-null	int64
attr_wifi	150243	non-null	int64
attr_good_for_kids	150243	non-null	int64
attr_restaurants_delivery	150243	non-null	int64
attr_outdoor_seating	150243	non-null	int64
attr_has_tv	150243	non-null	int64
attr_restaurants_reservations	150243	non-null	int64
attr_restaurants_good_for_groups	150243	non-null	int64
attr_alcohol	150243	non-null	int64
attr_by_appointment_only	150243	non-null	int64
attr_caters	150243	non-null	int64
attr_wheelchair_accessible	150243	non-null	int64
attr_good_for_meal	150243	non-null	int64
attr_restaurants_table_service	150243	non-null	int64
attr_dogs_allowed	150243	non-null	int64
attr_business_accepts_bitcoin	150243	non-null	int64
attr_happy_hour	150243	non-null	int64
attr_music	150243	non-null	int64
attr_drive_thru	150243	non-null	int64
attr_accepts_insurance	150243	non-null	int64
attr_coat_check	150243	non-null	int64
attr_good_for_dancing	150243	non-null	int64
attr_smoking	150243	non-null	int64
attr_byob	150243	non-null	int64
attr_corkage	150243	non-null	int64
attr_byob_corkage	150243	non-null	int64

attr_ages_allowed	150243	non-null	int64
attr_open_24_hours	150243	non-null	int64
attr_dietary_restrictions	150243	non-null	int64
attr_restaurants_counter_service	150243	non-null	int64

Table 2.1 Data specifications of all columns in Yelp Data set